CHE 565 -- Homework 3

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Problem 1

Steepest Descent

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At x0 =Matrix([[1], [1]]), f(x0) = 0.500000000000000

Gradient of f:

$$\nabla f = \left[\begin{pmatrix} 1.0x_1 + 1.0x_2 - 1.0 \\ 1.0x_1 + 2.0x_2 - 1 \end{pmatrix} \right]$$

At x0 =Matrix([[1], [1]]), Df(x0) = 1.00, 2.00

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 - 1.0\alpha \\ 1 - 2.0\alpha \end{bmatrix}$$

Function phi_k:

$$\phi(\alpha) = 6.5\alpha^2 - 5.0\alpha + 0.5$$

Derivative of phi_k:

$$\frac{d}{d\alpha}\phi(\alpha) = 13.0\alpha - 5.0$$
$$0 = 13.0\alpha - 5.0$$

Optimal alpha: 0.38

Next iterate x1: 0.62, x2: 0.23

Function value at next iterate: -0.46

Problem 2

Conjugate Gradient Method

$$\begin{bmatrix} s_1(\alpha) \\ s_2(\alpha) \end{bmatrix} = \begin{bmatrix} 0.0946745562130178 \\ -0.195266272189349 \end{bmatrix}$$

Problem 3

Constituents	Maximum Quantity (bbl/day)	Production Cost ($/bbl)
1	3000	26.00
2	2000	30.60
3	4000	29.20
4	1000	29.80

Grade	Specifications	Selling Price ($/bbl)
A	No more than 15% of 1; No more than 40% of 2	32.40
B	No more than 50% of 3; No more than 10% of 1	31.50
C	No less than 10% of 2; No more than 20% of 1	30.60

$$[A \leq 0.15c_1 \wedge A \leq 0.4c_2,\ B \leq 0.1c_1 \wedge B \leq 0.5c_3,\ C \geq 0.1c_2 \wedge C \leq 0.2c_1]$$